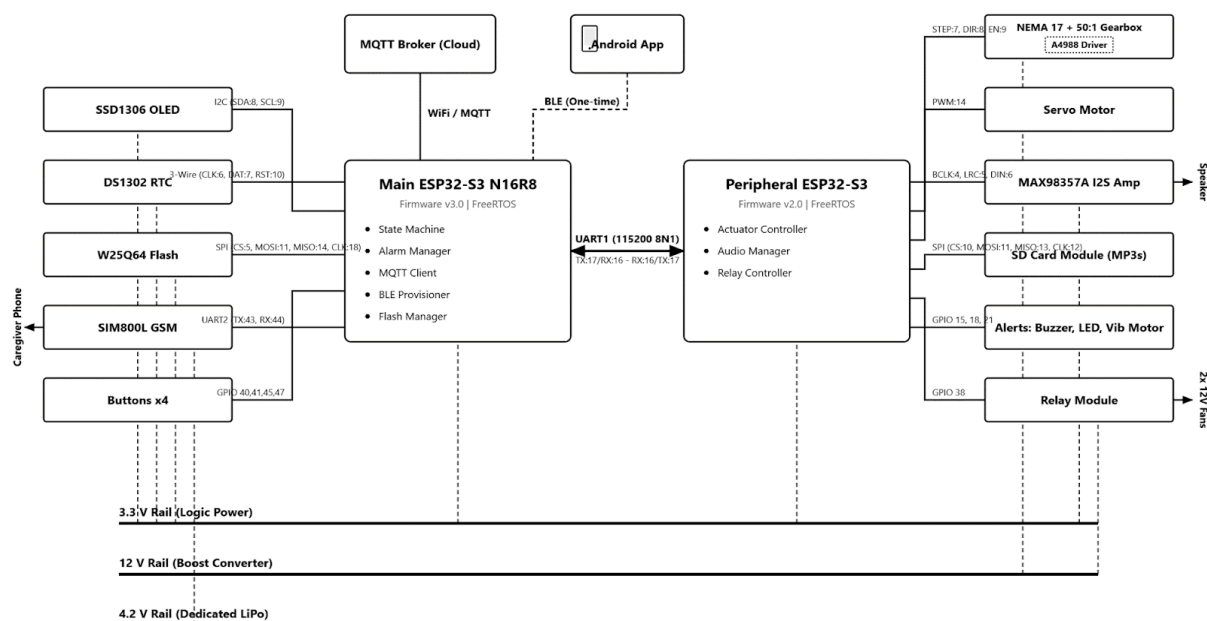


## 3.2 System Architecture

### 3.2.1 System Block Diagram

The system architecture is partitioned into five primary subsystems: the Main Control Unit (ESP32-S3 N16R8), the Peripheral Control Unit (ESP32-S3), the Dispenser Hardware Assembly, and the Android Companion Application. The two ESP32-S3 modules communicate via a 115,200-baud UART1 link; both connect to external hardware over I<sup>2</sup>C, SPI, and GPIO; and the main unit communicates wirelessly over BLE 5.0 and WiFi/MQT



**FIGURE 3.1 System Block Diagram**

The dual-ESP32-S3 architecture partitions responsibilities by domain: the main controller handles time-sensitive network operations and state management, while the peripheral controller executes all real-time actuator control and audio streaming. This purpose-built task partitioning is superior to the incidental dual-board designs in Works 3 and 5 [8,10], where separate chips were used solely to add WiFi capability to a legacy Arduino platform — resulting in increased integration complexity without architectural benefit.

### 3.2.2 System Performance Constraints

The following quantitative performance targets were derived from the SMART objectives of Section 1.3. These constitute the acceptance criteria for Chapter 4 verification activities.

Table 3.2: System Performance Constraints

| ID    | Parameter                                       | Target                           | Rationale                                                           | Design Element                                      |
|-------|-------------------------------------------------|----------------------------------|---------------------------------------------------------------------|-----------------------------------------------------|
| PC-01 | Alarm trigger latency (RTC match → alert onset) | $\leq 2$ s                       | One missed polling cycle (1 s) is clinically acceptable; two is not | 1 s RTC polling; synchronous handleAlarm() dispatch |
| PC-02 | Stepper travel time (max 20 stops)              | $\leq 8$ s                       | Must complete before hatch opens to prevent misalignment            | 10,000 steps/rev; max speed 1,000 steps/s           |
| PC-03 | BLE provisioning completion                     | $\leq 60$ s                      | User experience threshold for first-time setup                      | GATT write; 15 s WiFi connect timeout               |
| PC-04 | MQTT command latency (LAN/WAN)                  | $\leq 5$ s / $\leq 15$ s         | Schedule commands must not appear stale                             | QoS 1; keepalive 60 s                               |
| PC-05 | SMS delivery from alarm trigger                 | $\leq 60$ s                      | Caregiver must be informed promptly for intervention                | SIM800L AT+CMGS; dispatched in handleAlarm()        |
| PC-06 | Hatch open duration                             | $10 \text{ s} \pm 0.5 \text{ s}$ | Sufficient for tablet retrieval; limits multi-compartment exposure  | Hardware timer PWM; SERVO_OPEN_DURATION_MS          |
| PC-07 | Event log write reliability                     | $\geq 99$ %                      | Integrity of medication audit trail is a safety requirement         | Sector-validity byte; NVS-backed schedule           |
| PC-08 | WDT recovery time                               | $\leq 60$ s                      | Device must self-recover without human intervention                 | Hardware watchdog; 60 s timeout                     |

### 3.2.3 Data Flow Architecture

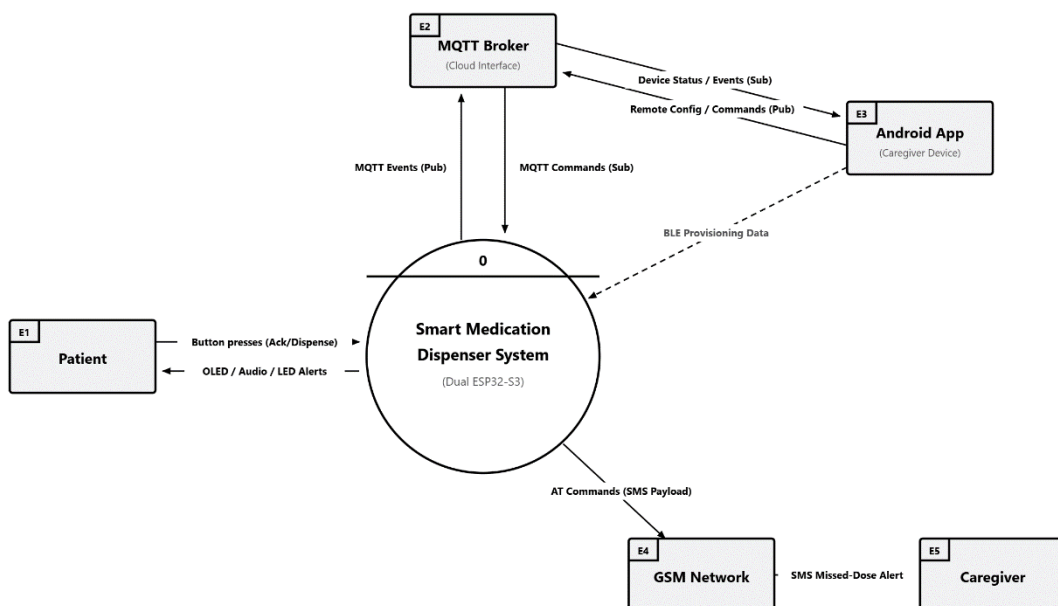
Three distinct data flows govern system communication, each with different latency, reliability, and persistence requirements.

**Provisioning Flow (Transient one-time):** Android application writes WiFi SSID, password, and MQTT broker address to BLE GATT Characteristic FFE2. Firmware validates credentials via WiFi association (15 s timeout). On success, Characteristic FFE3 notifies the

application, BLE advertising halts permanently, and credentials are committed to NVS. This flow carries security-sensitive data; its security analysis is in Section 3.5.1.

**Operational Control Flow (Periodic ongoing):** Schedule data is published from the Android application to `meddispenser/{deviceId}/schedule` at MQTT QoS 1. The device persists the received schedule to NVS immediately upon validated receipt. A status heartbeat is published by the device every 30 seconds to `meddispenser/{deviceId}/status`.

**Event Notification Flow (Event driven):** Alarm, dispensed, skipped, and missed-dose events are published from the device to `meddispenser/{deviceId}/alarm` at QoS 1. The application raises local Android push notifications and simultaneously records events in `localStorage`-persisted history. The GSM subsystem dispatches a parallel SMS notification, providing notification redundancy independent of internet connectivity.



**FIGURE 3.2 Context-Level (Level-0) Data Flow Diagram**

### 3.2.4 Power Distribution Architecture

Three voltage rails are derived from one 5 V USB-C input:

- **3.3 V Logic Rail:** Onboard LDO of each ESP32-S3 board; supplies both ESP32-S3 modules, SSD1306, DS1302, W25Q64, MAX98357A logic. Peak current ~500 mA.
- **12 V Motor Rail:** Step-up boost converter; supplies NEMA 17 via A4988 driver and the two 12 V DC cooling fans via the relay module. Electrically isolated; driver logic interface operates at 3.3 V.

- 4.0–4.4 V GSM Rail: LiPo cell (3.7 V nominal / 4.2 V charged); supplies SIM800L, which peaks at 2 A during burst transmission. Bulk decoupling: 100  $\mu$ F electrolytic || 100 nF ceramic at SIM800L supply pins.

*Design constraint:* GPIO 19/20 are reserved for native USB and must carry no external connections. GPIO 35–37 are internally connected to PSRAM and are similarly unavailable for peripheral use.

### 3.3 Hardware Design

#### 3.3.1 Component Selection Justification

Each major hardware component was selected through a structured evaluation against the performance constraints established in Section 3.2.2 and the functional requirements of Section 1.3. Table 3.3 documents the selection rationale for key components and identifies the alternatives that were systematically excluded.

Table 3.3: Hardware Component Selection Rationale

| Component Role                 | Selected                                           | Alternatives Considered                                                      | Justification for Selection                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------|----------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Central Microcontroller (Main) | ESP32-S3 N16R8                                     | Arduino Nano + ESP8266 [8]; Raspberry Pi Pico W [9]; Arduino + ESP32 [10]    | Dual-core 240 MHz with integrated BLE 5.0 and WiFi 4; 16 MB Flash + 8 MB PSRAM. FreeRTOS embedded in ESP-IDF provides preemptive multitasking, directly satisfying PC-01 ( $\leq 2$ s trigger latency) [12]. Paired with a dedicated peripheral ESP32-S3 for actuator and audio control.                                                                                 |
| Peripheral Microcontroller     | ESP32-S3 (any variant)                             | Single integrated ESP32-S3; standalone audio module                          | A dedicated second ESP32-S3 offloads all real-time actuator control (stepper, servo, buzzer, LED, vibration, relay) and MP3 audio from the network-intensive main controller, preventing network interrupt jitter from disturbing step-pulse timing. This purpose-built separation is superior to the legacy dual-chip approaches in Works 3 and 5 [8,10].               |
| Dispensing Actuator            | NEMA 17 Stepper + 50:1 Gearbox + A4988 (full-step) | DC motor (no position feedback); SG90 servo 180° rotation [7]; solenoid [10] | Stepper provides deterministic positional control across 21 discrete carousel stops without a closed-loop encoder [8]. The 50:1 planetary gearbox yields 10,000 output steps/revolution at 0.036°/step — sufficient resolution for the 21-stop carousel without micro-stepping. Full-step mode simplifies A4988 configuration and eliminates half-step resonance issues. |
| Hatch Actuator                 | Standard hobby servo (PWM)                         | Solenoid; DC motor; stepper motor                                            | Binary open/close motion of the dispensing hatch requires only two angular positions (0° closed, 90° open); a servo is the most cost-effective and mechanically reliable solution for this binary actuation role. The                                                                                                                                                    |

|                        |                                |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------|--------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |                                |                                                           | 10-second timed-close requirement (PC-06) maps naturally to a PWM pulse-width change after a software timer.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Wireless Communication | ESP32-S3 integrated WiFi + BLE | External WiFi module (ESP8266); separate BLE module       | On-chip integration eliminates inter-module SPI/UART communication overhead and removes one source of hardware failure. Bluetooth 5.0 provides extended range and improved connection stability over Bluetooth 4.2 present in older modules [17].                                                                                                                                                                                                                                                                                           |
| Cloud Protocol         | MQTT (HiveMQ broker)           | Firebase Realtime Database [7]; HTTP REST API; WebSockets | MQTT's publish-subscribe model is bandwidth-efficient and designed for constrained embedded clients; the lightweight PubSubClient library (RAM footprint ~2 KB) is well-suited to the ESP32-S3's SRAM constraints. Firebase introduces vendor lock-in and per-connection costs, as identified as a limitation of Work 2 [7]. HTTP REST APIs are request-response only, preventing event-push from device to application. QoS 1 delivery semantics satisfy PC-04 with guaranteed delivery confirmation [18].                                 |
| Timekeeping            | DS1302 RTC                     | DS3231 RTC; internal ESP32 RTC                            | DS1302 requires only a 3-wire interface (CLK, DAT, RST), conserving the limited GPIO budget. CR2032 backup cell sustains timekeeping during power removal (300 nA backup current). The $\pm 2$ ppm accuracy is adequate for alarm applications when periodic NTP synchronisation is applied via the sync_time MQTT command [15]. DS3231 offers higher accuracy but requires I <sup>2</sup> C, conflicting with the OLED for bus address management.                                                                                         |
| Event Log Storage      | W25Q64 SPI NOR Flash           | MicroSD card; internal ESP32-S3 NVS Flash                 | W25Q64 provides deterministic SPI initialisation without SD card detection protocols, directly resolving the voltage-transient-induced initialisation failures encountered during Iteration 1 (Section 3.1.2). NOR Flash offers byte-level read access suitable for structured records. 100,000 program/erase cycles per 4 KB sector [16] far exceeds the projected operational lifetime at one event per dispensing per day. Internal NVS Flash was reserved for firmware and configuration data to avoid wear from frequent event writes. |
| Caregiver Notification | SIM800L GSM                    | No fallback; LoRa; NB-IoT                                 | GSM/SMS coverage is universal across Ghana's major carriers (MTN, Vodafone, AirtelTigo), as opposed to LoRa (requires                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                      |                                |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|--------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      |                                |                                                                            | gateway infrastructure) and NB-IoT (limited Ghanaian coverage in 2026). SMS delivery is network-confirmed and does not require the recipient to have a smartphone. This satisfies O4 and PC-05 and directly addresses the GSM fallback gap identified in all six related works [7,8,9,10,11].                                                                                                                                                                                                                                                                             |
| Mobile App Framework | React + Capacitor (TypeScript) | Native Android (Java/Kotlin); Flutter; MIT App Inventor [7]; Ionic Angular | React's component model and virtual DOM enable rapid iterative UI development with type-safe TypeScript. Capacitor provides a typed native plugin bridge to BLE (via @capacitor-community/bluetooth-le) and Android notifications, supporting all required native APIs without a full Java rewrite. MIT App Inventor (used in Work 2) is feature-limited and unsuitable for the complex BLE GATT and MQTT interactions required [7]. Native Kotlin would provide optimal performance but requires maintaining separate codebases for iOS, which is outside project scope. |

### 3.3.2 Bill of Materials

Table 3.4: Bill of Materials Smart Medication Dispensing Machine

| #  | Component                                   | Specification                                                 | Qty | Unit Cost (GHS) | Total (GHS) |
|----|---------------------------------------------|---------------------------------------------------------------|-----|-----------------|-------------|
| 1  | ESP32-S3 N16R8 Dev Board (Main MCU)         | 240 MHz dual-core, BLE 5.0, WiFi 4, 16 MB Flash, 8 MB PSRAM   | 1   | 120             | 120         |
| 2  | ESP32-S3 Dev Board (Peripheral MCU)         | ESP32-S3 variant; actuator + audio controller (Firmware v2.0) | 1   | 120             | 120         |
| 3  | NEMA 17 Stepper Motor                       | 42×42 mm, 1.8°/step, 0.9 A, 40 N·cm                           | 1   | 100             | 100         |
| 4  | A4988 Stepper Driver                        | Full-step mode; 2 A max; STEP/DIR interface                   | 1   | 25              | 25          |
| 4a | 50:1 Planetary Gearbox (NEMA 17-compatible) | 50:1 reduction; suits NEMA 17 motor output shaft              | 1   | 100             | 100         |
| 5  | Servo Motor                                 | Standard hobby, 50 Hz PWM, 0–180°                             | 1   | 30              | 30          |
| 6  | DS1302 RTC Module                           | 3-wire, CR2032, ±2 ppm                                        | 1   | 20              | 20          |
| 7  | CR2032 Coin Cell                            | 3 V lithium, RTC backup                                       | 1   | 5               | 5           |

|    |                                      |                                                           |       |          |           |
|----|--------------------------------------|-----------------------------------------------------------|-------|----------|-----------|
| 8  | W25Q64 NOR Flash (SOIC-8)            | 64 Mbit / 8 MB, SPI up to 104 MHz, 3.3 V                  | 1     | 30       | 30        |
| 9  | SSD1306 OLED Module                  | 128×64 px, I <sup>2</sup> C 0x3C, 0.96 inch               | 1     | 30       | 30        |
| 10 | SIM800L GSM/GPRS Module              | Quad-band 850/900/1800/1900 MHz, AT cmds                  | 1     | 85       | 85        |
| 11 | SIM Card (prepaid)                   | MTN/ /AirtelTigo, active                                  | 1     | 15       | 15        |
| 12 | MAX98357A I <sup>2</sup> S Amplifier | 3 W mono class-D, I <sup>2</sup> S, 9 dB gain             | 1     | 25       | 25        |
| 13 | MicroSD Card Module (SPI)            | FAT32, for audio ESP32                                    | 1     | 15       | 15        |
| 14 | MicroSD Card ≥ 4 GB                  | FAT32 Class 10; MP3 files                                 | 1     | 45       | 45        |
| 15 | Speaker (8 Ω, 0.5 W)                 | Compact, alarm audio output                               | 1     | 20       | 20        |
| 16 | Active Buzzer (5 V)                  | HIGH = ON; secondary alert                                | 1     | 8        | 8         |
| 17 | Status LED (5 mm, green)             | Via 220 Ω series resistor                                 | 1     | 2        | 2         |
| 18 | Vibration Motor Module               | ERM DC, 3–5 V, GPIO driven                                | 1     | 10       | 10        |
| 19 | Relay Module (1-channel, 5 V)        | Controls 12 V load; GPIO-triggered; controls 2× 12 V fans | 1     | 20       | 20        |
| 20 | 12 V DC Cooling Fan (80 mm)          | Enclosure/driver cooling; relay-switched                  | 2     | 45       | 90        |
| 21 | Tactile Push Button (6×6 mm)         | Momentary PCB-mount; ×4 (main MCU)                        | 4     | 2        | 8         |
| 22 | Boost DC-DC Converter (12 V)         | For stepper motor + cooling fan power rail                | 1     | 20       | 20        |
| 23 | Lithium battery Cell 3.7 V,          | Power supply                                              | 4     | 35       | 35        |
| 24 | 12v/ 3 A USB-C Power Adapter         | Main system supply                                        | 1     | 100      | 100       |
| 25 | 3D-Printed Carousel (PLA)            | 21-compartment rotating; custom                           | 1     | 600      | 600       |
| 26 | 3D-Printed Enclosure (PLA)           | Main housing; hatch aperture                              | 1     | 700      | 700       |
| 27 | Jumper Wires, Stripboard, Headers    | Prototype interconnect                                    | 1 lot | 45       | 45        |
| 28 | Decoupling Capacitors                | 100 μF electrolytic + 100 nF ceramic                      | 1 lot | 5        | 5         |
| 29 | Resistors                            | 220 Ω (LED), 10 kΩ (pull-up)                              | 1 lot | 3        | 3         |
|    |                                      |                                                           |       | Subtotal | 2,676 GHS |

### 3.3.3 GPIO Allocation and Pin Assignment

Because the system uses two ESP32-S3 microcontrollers, the GPIO budget is presented as two separate tables one for each unit.

Table 3.5a: Main ESP32-S3 N16R8 GPIO Allocation (Firmware v3.0)

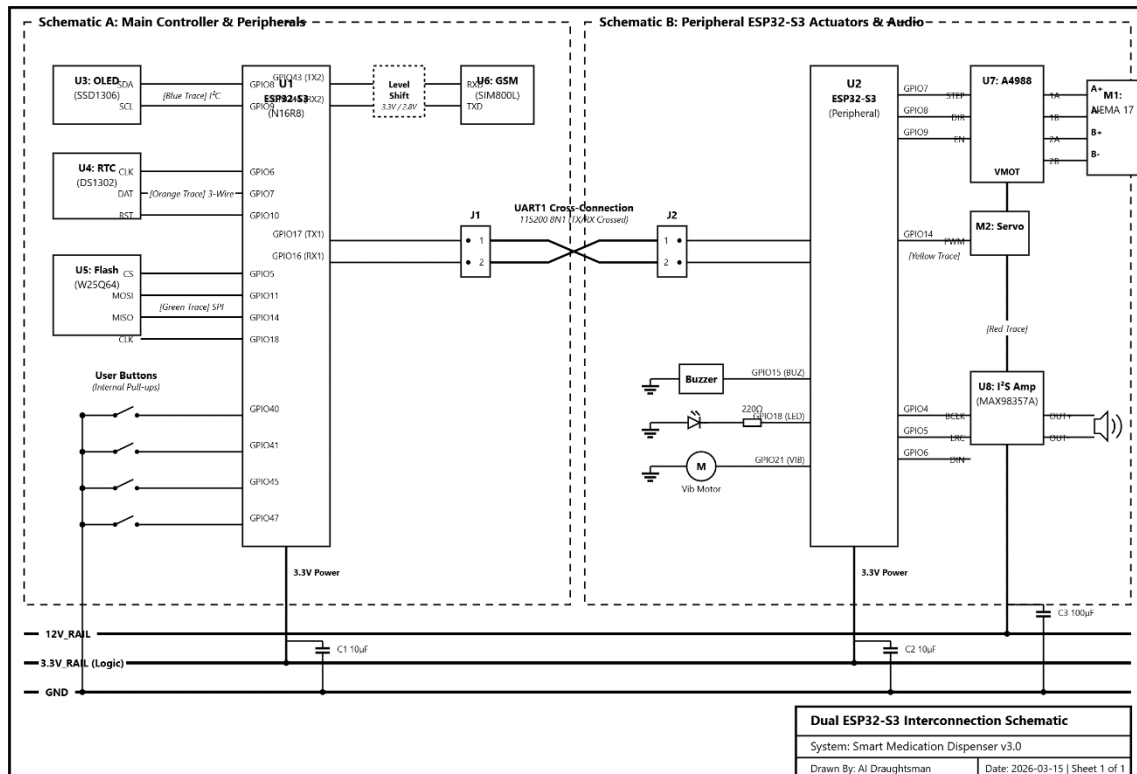
| PIO   | Function   | Connected Peripheral   | Interface              | Notes                                          |
|-------|------------|------------------------|------------------------|------------------------------------------------|
| 5     | FLASH_CS   | W25Q64 pin 1           | SPI CS                 | Active LOW                                     |
| 6     | RTC_CLK    | DS1302 CLK             | 3-wire clock           |                                                |
| 7     | RTC_DAT    | DS1302 DAT/IO          | 3-wire data            | Bidirectional                                  |
| 8     | I2C_SDA    | SSD1306 SDA            | I <sup>2</sup> C data  |                                                |
| 9     | I2C_SCL    | SSD1306 SCL            | I <sup>2</sup> C clock |                                                |
| 10    | RTC_RST    | DS1302 RST/CE          | 3-wire CE              |                                                |
| 11    | FLASH_MOSI | W25Q64 DI (pin 5)      | SPI MOSI               |                                                |
| 14    | FLASH_MISO | W25Q64 DO (pin 2)      | SPI MISO               |                                                |
| 16    | PERIPH_RX  | Peripheral ESP32-S3 TX | UART1 RX               | 115,200 baud                                   |
| 17    | PERIPH_TX  | Peripheral ESP32-S3 RX | UART1 TX               | 115,200 baud                                   |
| 18    | FLASH_CLK  | W25Q64 CLK (pin 6)     | SPI CLK                | 10 MHz                                         |
| 19    | RESERVED   | —                      | Native USB             | Must not be connected                          |
| 20    | RESERVED   | —                      | Native USB             | Must not be connected                          |
| 35–37 | RESERVED   | —                      | Octal PSRAM            | Not user-accessible                            |
| 40    | BTN_BACK   | Button → GND           | Digital IN             | Internal pull-up                               |
| 41    | BTN_RESET  | Button → GND           | Digital IN             | Internal pull-up; 5 s long-press factory reset |
| 43    | GSM_TX     | SIM800L RX             | UART2 TX               | 9600 baud                                      |
| 44    | GSM_RX     | SIM800L TX             | UART2 RX               | 9600 baud                                      |
| 45    | BTN_NAV    | Button → GND           | Digital IN             | Internal pull-up                               |
| 47    | BTN_SELECT | Button → GND           | Digital IN             | Internal pull-up                               |

Table 3.5b: Peripheral ESP32-S3 GPIO Allocation (Firmware v2.0)

| GPI O | Function  | Connected Peripheral    | Interface   | Notes                   |
|-------|-----------|-------------------------|-------------|-------------------------|
| 2     | BUZZER    | Active Buzzer           | Digital OUT | HIGH = ON               |
| 4     | SERVO_PWM | Servo Motor signal wire | PWM 50 Hz   | 1000–2000 $\mu$ s pulse |
| 12    | VIB_MOTOR | Vibration Motor         | Digital OUT | HIGH = ON               |



|    |            |                          |                              |                                  |
|----|------------|--------------------------|------------------------------|----------------------------------|
| 13 | STEP_EN    | A4988 EN                 | Digital OUT                  | Active LOW (enabled)             |
| 15 | STATUS_LED | LED + 220 $\Omega$ → GND | Digital OUT                  |                                  |
| 16 | PERIPH_RX  | Main ESP32-S3 TX         | UART1 RX                     | 115,200 baud                     |
| 17 | PERIPH_TX  | Main ESP32-S3 RX         | UART1 TX                     | 115,200 baud                     |
| 22 | I2S_DOUT   | MAX98357A DIN            | I <sup>2</sup> S data        |                                  |
| 25 | I2S_LRC    | MAX98357A LRC            | I <sup>2</sup> S word select |                                  |
| 26 | I2S_BCLK   | MAX98357A BCLK           | I <sup>2</sup> S bit clock   |                                  |
| 34 | SD_CS      | MicroSD Card CS          | SPI CS                       |                                  |
| 35 | SD_MOSI    | MicroSD Card DI          | SPI MOSI                     |                                  |
| 36 | SD_MISO    | MicroSD Card DO          | SPI MISO                     |                                  |
| 37 | SD_CLK     | MicroSD Card CLK         | SPI CLK                      |                                  |
| 38 | RELAY      | Relay Module IN          | Digital OUT                  | HIGH = Fans ON; 20-min cycle     |
| 39 | STEP_DIR   | A4988 DIR                | Digital OUT                  |                                  |
| 41 | STEP_STEP  | A4988 STEP               | Digital OUT                  | Step pulse; Core 0 FreeRTOS task |



***FIGURE 3.3 Schematic Interconnection Diagrams***